

Dialogue Summarization Using Pre-trained Language Models on SAMSum

COMP7045 Natural Language Processing & Large Language Model

Naibo Yu



香港浸會大學
HONG KONG BAPTIST UNIVERSITY



DEPARTMENT OF
COMPUTER SCIENCE
計算機科學系

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Background & Motivation

Why Dialogue Summarization?

Applications

- Customer service chat logs
- Meeting & call transcripts
- Social media conversations

Unique Challenges

- Multi-speaker turn structure
- Informal language & emoji
- Topic drift across turns
- Implicit co-references

From Rules to Transformers

- 1958 Luhn — word frequency heuristics
- 2015 Rush et al. — first neural model
- 2017 See et al. — Pointer-Generator
- 2019 BART · T5 — seq2seq pretraining
- 2019 SAMSum dataset introduced
- 2020 PEGASUS — gap-sentence pretraining
- 2023+ LLMs (GPT-4) zero-shot summarization

SAMSum Corpus

16,369

Total dialogues

78.3%

Compression ratio

93.8

Avg. words / dialogue

Split	Samples	Avg. Dialogue (words)	Avg. Summary (words)	Max Dialogue (words)
Train	14,731	93.8	20.3	803
Validation	818	91.6	20.3	540
Test	819	95.5	20.0	516

Two Approaches: Extractive vs. Abstractive

① TextRank (Baseline)

Type: Extractive — copies sentences verbatim

Algorithm: Graph-based PageRank on sentence similarity

Library: Sumy (Python)

Output: Top-2 ranked sentences concatenated

- ✓ No training required
- ✓ Fast & interpretable
- ✗ Cannot paraphrase
- ✗ Fails on fragmented turns

② BART Fine-tuning (Our Model)

Base: facebook/bart-large-cnn (406.3M params)

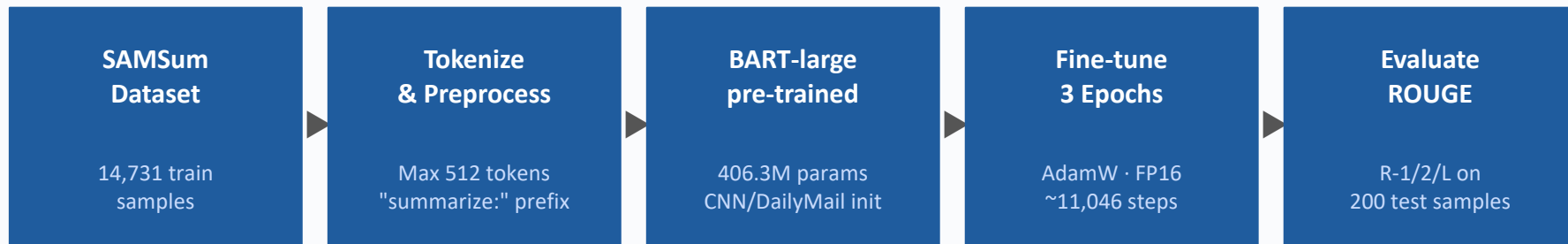
Type: Abstractive — generates new text

Epochs: 3 · LR: 5×10^{-5} · Batch: 4

Input: "summarize: " + dialogue (max 512 tokens)

- ✓ Captures cross-turn semantics
- ✓ Fluent paraphrased output
- ✗ Risk of hallucination
- ✗ Truncation on long dialogues

BART Fine-tuning Pipeline



Key Hyperparameters

Learning Rate	Batch Size	Warmup Steps	Max Input Len	Eval Metric	Precision
5×10^{-5}	4 / 8	200	512 tokens	ROUGE-L	FP16 (GPU)

ROUGE Evaluation Results

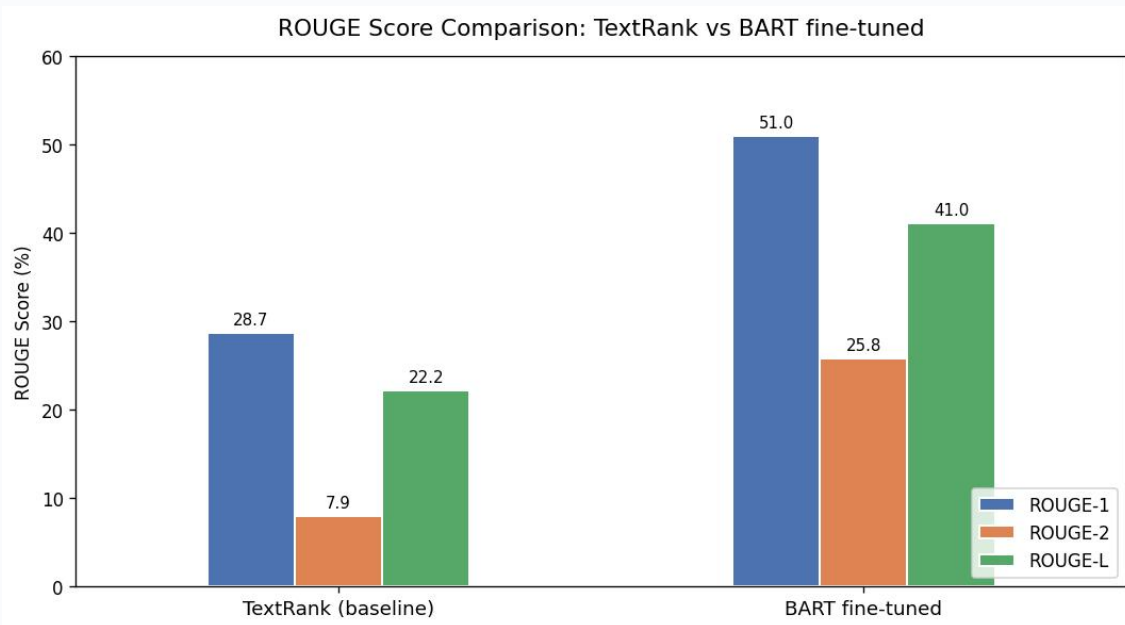


Figure 1: ROUGE-1, ROUGE-2, and ROUGE-L comparison between TextRank and BART fine-tuned.

+22.31

+77.7% relative

ROUGE-1

+17.86

+224.9% relative

ROUGE-2

+18.84

+84.9% relative

ROUGE-L

Per-sample ROUGE-L Distribution

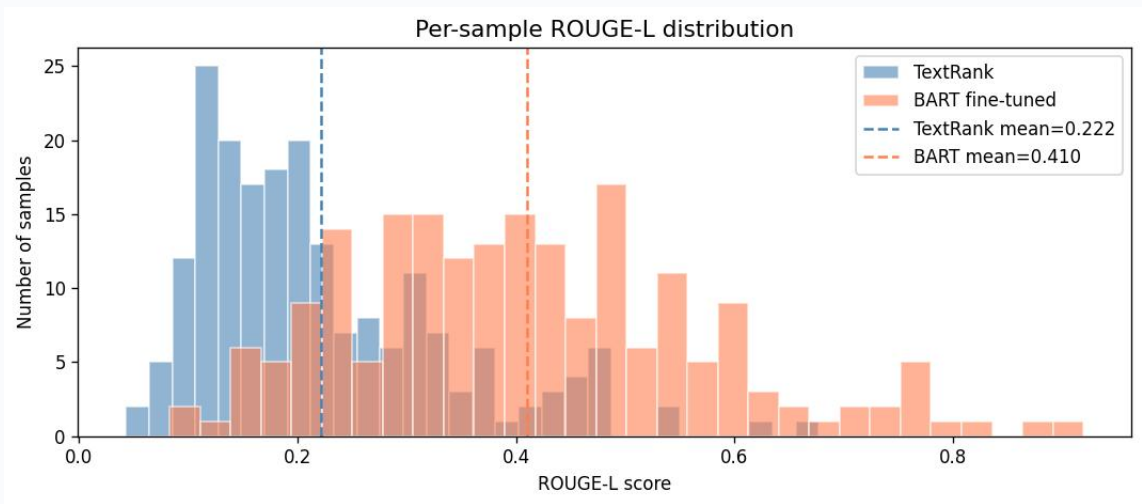


Figure 2: Per-sample ROUGE-L distributions. TextRank (blue) vs BART (coral). Dashed lines = means.

Key Observations

- BART distribution shifts markedly right vs TextRank
- Lower variance — more consistent across samples
- TextRank heavy left tail: near-zero scores on short dialogues
- BART mean ROUGE-L ≈ 0.41
TextRank mean ≈ 0.22
- **ROUGE-2 gain is largest (+3.25×)**

Qualitative Analysis: Three Cases

Case #70 — Best Case

Dialogue: "Ali: I left my wallet... Mohammad: Found it! Bring it to uni tomorrow? — Sure thing."

Reference: Ali left his wallet at Mohammad's. Mohammad'll bring it to uni tomorrow.

TextRank: Give me a sec... Phew, I don't know what I'd do if it wasn't there.

BART: Ali left his wallet yesterday. Mohammad found it and will bring it to university tomorrow.

✓ BART captures outcome correctly

Case #5 — Typical Case

Dialogue: "Hilary: I've got the keys. Benjamin: I'll pass by at lunchtime and take a nap. Hilary: La Cantina, 2pm..."

Reference: Hilary has keys. Benjamin will get them at lunch. Hilary meets French people at 2pm.

TextRank: "Yeah, I guess so... I'm with Hilary atm... take the keys and go take a nap..."

BART: Hilary has got the keys. She meets French people at La Cantina at 2pm. Benjamin takes a nap.

✓ BART fluent · minor hallucination ("Italian cuisine")

Case #140 — Failure Case

Dialogue: "Sean: I overslept :/ Sam: Again?? Sean: I know."

Reference: Sean overslept again.

TextRank: Sean: I overslept :/ Sam: Again?? Sean: I know.

BART: Sean overslept again and Sam is angry with him. He's not happy.

✗ BART hallucinates emotions — TextRank wins on ROUGE-L

Conclusions & Future Work

01

Strong Quantitative Gains

BART achieves ROUGE-1 51.03 vs TextRank 28.72 (+77.7%). ROUGE-2 improves 3.25×, confirming abstractive superiority. Results consistent with published SAMSum SOTA (~52–53).

02

Cross-turn Semantics Matter

TextRank fundamentally fails when dialogue meaning is distributed across turns. BART synthesizes multi-speaker content into fluent, coherent summaries that extractive methods cannot produce.

03

Hallucination on Short Dialogues

BART over-generates on very short inputs (< 20 words), inferring unsupported emotional states. Extractive methods remain safer for minimal-context dialogues.

Future Work: Long-document handling (Longformer/LED) · Faithfulness metrics beyond ROUGE · Domain adaptation for medical & legal dialogues